

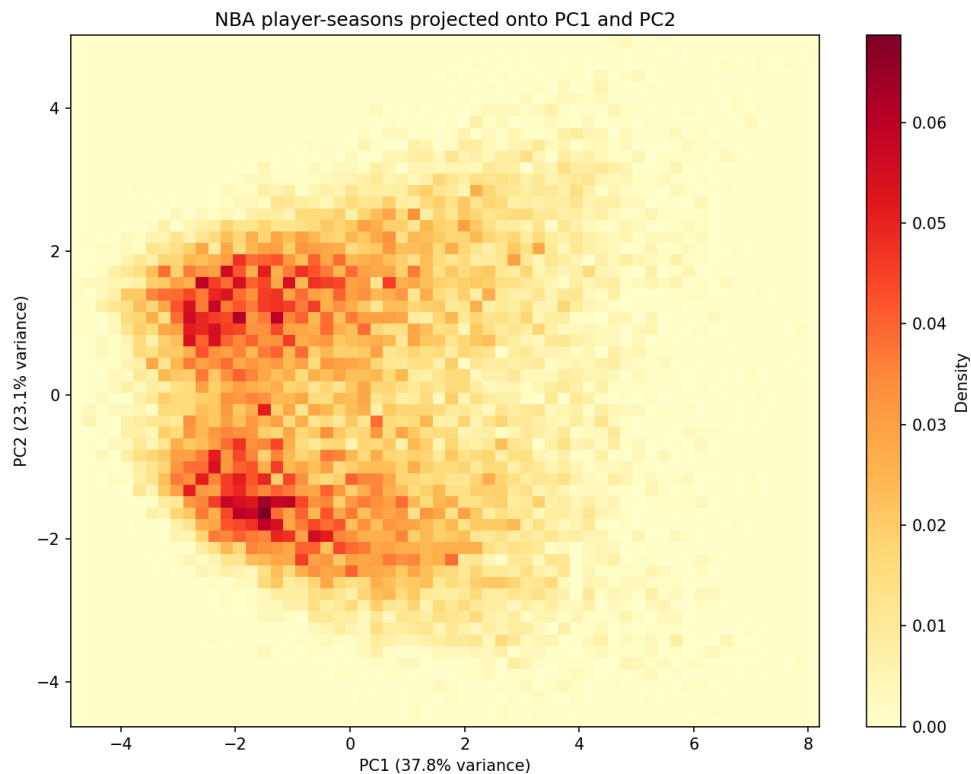
Who is the NBA GOAT? A dimension reduction analysis of the NBA since 1947

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NBA Player-Seasons: PCA Projection

13 per-game statistics standardized and projected onto two principal components



Stat	PC1	PC2
points	0.408	0.067
avg_minutes	0.402	0.032
freeThrowsAttempted	0.390	-0.082
freeThrowsPercentage	0.370	0.008
turnovers	0.298	0.200
assists	0.250	0.272
reboundsTotal	0.249	-0.333
fieldGoalsPercentage	0.223	-0.235
steals	0.218	0.272
foulsPersonal	0.214	-0.327
blocks	0.143	-0.253
threePointersAttempted	0.069	0.471
threePointersPercentage	0.022	0.490

PC1 = 37.8% variance · PC2 = 23.1% variance · Total = 60.9%

Interpreting the Principal Components

PCA on 13 per-game stats across all NBA player-seasons · PC1 explains 37.8% of variance, PC2 explains 23.1%

PC1

Overall production / usage

37.8% of variance

Captures how much a player does — a general star / workhorse axis.

High PC1

High-volume, high-usage player (stars, workhorses)

Low PC1

Low-minute bench players with minimal statistical footprint

Key stats

Points, avg. minutes, free throw attempts, turnovers

PC2

Perimeter vs. big-man style

23.1% of variance

Captures style of play, not volume — a guard/big positional axis.

High PC2

Guard-like: 3PA, 3P%, assists, steals

Low PC2

Big-man: blocks, rebounds, fouls, low FG%

Key insight

Orthogonal to PC1 — style is independent of usage

Together PC1 and PC2 explain 60.9% of total variance in the dataset

High PC1 — Volume Stars

Player-seasons with the highest overall usage and statistical output

Player	Year	PC1	PC2
James Harden	2019	8.20	4.81
Michael Jordan	1987	7.85	0.84
Wilt Chamberlain	1962	7.83	-3.79
David Robinson	1994	7.54	-2.19
Charles Barkley	1986	7.42	-1.23
Joel Embiid	2022	7.15	0.02
Hakeem Olajuwon	1993	7.05	-3.25
Giannis Antetokounmpo	2022	7.01	-0.28
Allen Iverson	2005	6.91	3.78
Russell Westbrook	2017	6.89	3.75

PC1 = usage/volume axis · PC2 = guard/big axis · Best season shown per player

Low PC1 — Bench / Low Usage (Who are these guys?)

Players with the lowest statistical volume across all categories

Player	Year	PC1	PC2
Gary Clark	2021	-4.87	1.61
Rashad Vaughn	2016	-4.82	1.47
Sidney Lowe	1990	-4.76	0.95
Damon Jones	2007	-4.59	2.18
Andy Phillip	1958	-4.58	-0.20
Hot Rod Williams	1999	-4.56	-0.35
Michael Cage	1998	-4.53	-0.40
Haywood Highsmith	2022	-4.52	1.09
Nicolas Batum	2026	-4.50	1.63
John Wetzel	1973	-4.48	-0.38

PC1 = usage/volume axis · PC2 = guard/big axis · Best season shown per player

High PC2 — Most Guard-Like

Players defined by three-point shooting and playmaking

Player	Year	PC1	PC2
Stephen Curry	2021	4.79	5.01
Trae Young	2024	4.88	4.99
Damian Lillard	2020	5.25	4.84
James Harden	2019	8.20	4.81
Michael Adams	1991	5.62	4.70
Mookie Blaylock	1996	2.17	4.63
Tyrese Haliburton	2023	3.63	4.47
Devonte' Graham	2020	1.76	4.45
LaMelo Ball	2024	5.14	4.42
Mark Price	1990	3.89	4.41

PC1 = usage/volume axis · PC2 = guard/big axis · Best season shown per player

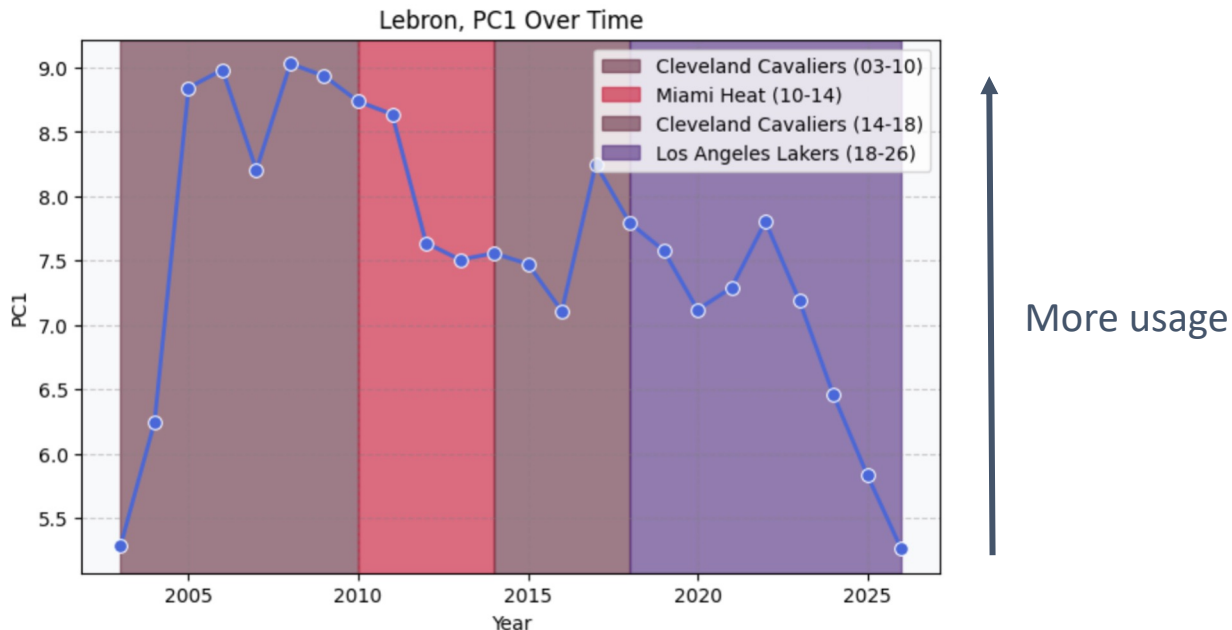
Low PC2 — Most Big-Like

Players defined by shot-blocking, rebounding, and interior play

Player	Year	PC1	PC2
Kareem Abdul-Jabbar	1975	6.08	-4.62
Dikembe Mutombo	1995	3.74	-4.61
DeAndre Jordan	2015	3.38	-4.58
Mark Eaton	1985	2.62	-4.52
Tree Rollins	1983	1.11	-4.50
Shaquille O'Neal	1992	6.25	-4.43
Hassan Whiteside	2015	1.88	-4.29
Artis Gilmore	1981	4.03	-4.27
Wilt Chamberlain	1959	6.33	-4.26
Rudy Gobert	2021	3.38	-4.22

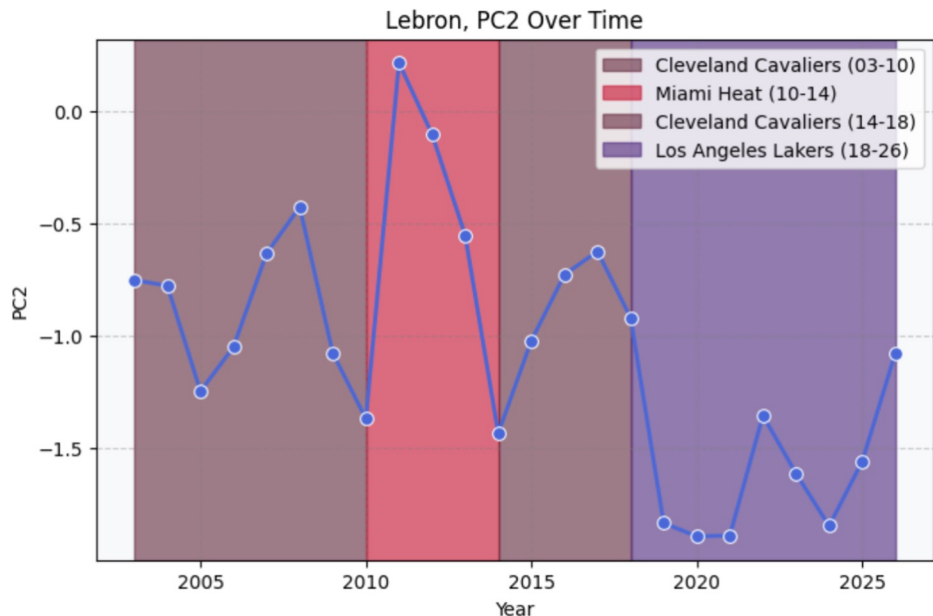
PC1 = usage/volume axis · PC2 = guard/big axis · Best season shown per player

Lebron James PC1 Over Time



- Usage highest in Cleveland at beginning of career, and in second stint after Kyrie leaves
- Usage drops as LeBron ages while playing for the Lakers

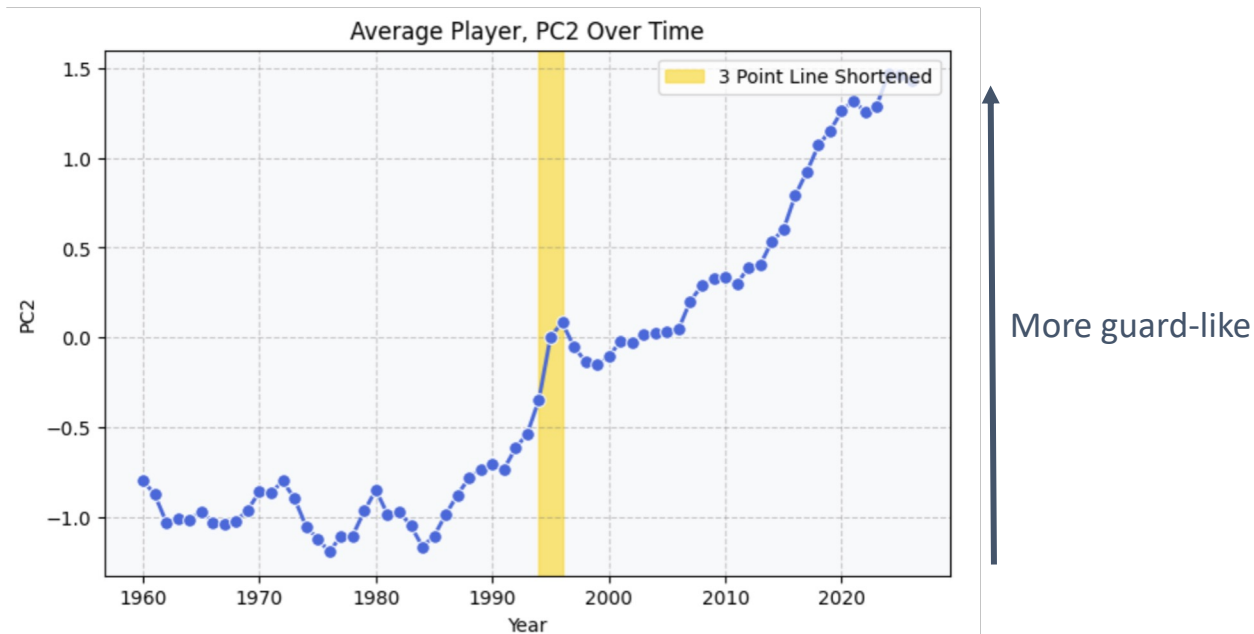
Lebron James PC2 Over Time



More center-like

- LeBron becomes more guardlike as career goes on
- Most center-like on Miami Heat

Average Player PC2 Over Time



- We use PCA on the whole dataset, rather than modern era
- Guard-like-ness jumps when 3-point line is shortened, and then falls when it is lengthened

Most Productive Versatile Players

Highest PC1 (usage/production) with PC2 between -1.5 and 1.5 (neither pure guard nor pure big)

Player	Year	PC1	PC2
Michael Jordan	1987	7.85	0.84
Charles Barkley	1986	7.42	-1.23
Joel Embiid	2022	7.15	0.02
Giannis Antetokounmpo	2022	7.01	-0.28
Karl Malone	1989	6.60	-1.10
Nikola Jokic	2022	6.51	0.78
Dwyane Wade	2007	6.37	1.25
DeMarcus Cousins	2017	6.37	1.04
Bernard King	1985	6.23	-0.70
Shai Gilgeous-Alexander	2023	6.12	1.47

PC1 = usage/volume axis · PC2 = guard/big axis · Best season shown per player

Career GOAT Score Using PC1 & PC2

Career PC1 + Versatility Bonus + Consistency Bonus + Rings Bonus

Player	Years	Rings	GOAT Score	Career PC1	PC1 “Sharpe”	Avg PC1	Consistency (PC1 STDEV)	Avg PC2
LeBron James	24	4	162.64	123.36	6.89	5.14	0.75	2.46
Michael Jordan	17	6	132.70	93.99	3.45	5.53	1.60	1.02
Kareem Abdul-Jabbar	21	6	117.38	85.05	2.44	4.05	1.66	-2.93
Shaquille O'Neal	19	4	116.35	91.66	3.01	4.82	1.60	-3.41
Kobe Bryant	20	5	112.52	84.69	2.19	4.24	1.93	2.02
Kevin Durant	19	2	109.71	90.80	4.97	4.78	0.96	1.87
Karl Malone	19	0	107.84	98.38	4.23	5.18	1.22	-1.28
Magic Johnson	14	5	100.31	72.45	6.55	5.17	0.79	1.60
Hakeem Olajuwon	19	2	100.03	87.61	1.87	4.61	2.46	-2.58
James Harden	18	0	92.17	89.72	1.99	4.99	2.50	3.40

GOAT Score = Career PC1 × Versatility Bonus (+5% max) × Consistency Bonus (+10% max) × Rings Bonus (+5% per ring)